Task 9: Time Series Analysis

As instructed, no time series model (trend decomposition, ARIMA, or forecast) was fitted for this task. However, our dataset does contain a date field (bill_date, 2014–2023), so we performed one small **exploratory check** of what this field shows - not to analyze it, but to ground this discussion in our own dataset's real limitations rather than purely hypothetical ones.

9.1 What Our Own Data Shows and Why It Is Not Suitable for Time Series Analysis

bill_date has 100% coverage (27,034/27,034 cases), spanning 21 April 2014 to 11 August 2023. On the surface, nearly a decade of dates looks like promising material for trend and seasonal analysis. Closer inspection reveals a fundamental problem:

Institution	Years present
Institution_A	2023 only
Institution_B	2015–2023 (near-continuous)
Institution_C	2014–2015 only
Institution_D	2018, 2020–2023 (missing 2014–2017, 2019)

Each institution contributes data during a different, largely non-overlapping period. This is not a minor caveat. It means a naive year-by-year pooled anemia prevalence figure is not tracking a genuine population trend at all; it is tracking which institutions happened to be contributing data that year. Our exploratory chart makes this concrete: pooling all institutions together shows prevalence swinging from 51.4% (2014) down to 29.8% (2017) and back up to 50.9% (2023) - a pattern that, on inspection, is substantially explained by Institution_C's temporary 2014–2015 presence (elevated baseline prevalence, per Task 4) rather than any real change in population health. Restricting to Institution_B only, the one institution with near-continuous coverage, tells a different and more coherent story: a comparatively steady rise from 28.7% (2015) to 62.6% (2023), which is at least internally consistent, though it comes from a single site and cannot be generalized to the other institutions or to Sri Lanka as a whole.

This finding is itself a useful, concrete lesson for the client: before any time series or forecasting exercise can be trusted, the underlying data collection must be continuous and consistent for the unit being tracked. Our dataset, assembled retrospectively from routine records across four institutions with uncoordinated collection periods, was never designed for this purpose - a limitation worth stating plainly rather than working around.

9.2 Trend Analysis

What it would involve: identifying the long-term underlying direction of anemia prevalence (or mean HB) over time, separate from short-term noise and seasonal fluctuation - typically via moving averages, linear/polynomial trend fitting, or the trend component of a formal decomposition (e.g., STL).

How it would apply here, if data allowed: a properly time-structured version of this dataset, ideally with consistent monthly reporting from all four institutions across the full period - would let the client ask whether anemia burden is genuinely rising, falling, or stable, and whether that trend differs by institution, age group, or sex. This is precisely the approach taken in recent published work: a 2024 study forecasting anemia prevalence among women of reproductive age in Ghana used two decades of consistent annual data to fit trend and ARIMA models, finding a slow decline projected to continue only marginally to 2030; a comparable study applied ARIMA to two decades of Iraqi data on anemia among women of reproductive age specifically because the outcome affects both maternal and child health and needed a forecast for future resource planning, precisely the same policy motivation as our own project's "resource planning" sub-topic.

Why our data cannot currently support this: as shown in 9.1, our institution-level discontinuities would confound any pooled trend estimate. Even the single-institution (Institution_B) series, while more internally consistent, spans annual case volumes ranging from 595 to 3,146 - itself suggesting inconsistent reporting practices or catchment changes over time, not a stable, comparable annual sample.

9.3 Seasonal Analysis

What it would involve: detecting recurring within-year patterns (e.g., monthly or quarterly cycles) separate from the long-term trend, relevant wherever a health outcome is plausibly linked to a cyclical driver such as climate, disease transmission seasons, or agricultural/nutritional cycles.

How it would apply here, if data allowed: there is strong published precedent for expecting genuine seasonality in hemoglobin and anemia-related outcomes in tropical and monsoon-affected settings, directly relevant to Sri Lanka's own monsoon climate. Research in West Africa has linked seasonal childhood anemia to the malaria transmission season, with mean hemoglobin levels measurably falling over the course of the malaria season. Broader reviews confirm that in malaria-endemic tropical regions, severe anemia requiring transfusion in young children is a major cause of hospital admission, particularly during the rainy season months when malaria transmission peaks. Closer to our own context, a study of malaria and hematological profiles in a South Asian refugee camp setting found peak transmission during the monsoon season (June–September), with infected individuals showing measurable anemia and other hematological alterations. If our dataset had consistent monthly reporting, we would specifically test whether HB levels or anemia prevalence dip during Sri Lanka's monsoon periods (the two monsoon seasons affecting the island at different times of year), which would have direct, actionable implications for **when** to schedule supplementation or screening campaigns.

Why our data cannot currently support this: with institutions contributing data in different, non-overlapping year ranges, we cannot even reliably construct a single continuous monthly series, let alone decompose it into trend and seasonal components with any confidence.

9.4 Forecasting

What it would involve: projecting future values of an outcome (e.g., anemia prevalence, screening test volume) based on patterns identified in historical data, used to anticipate future resource needs rather than only describing the past.

How it would apply here, if data allowed: forecasting is precisely where this project's "resource planning" sub-topic would be most directly realized. Recent applications illustrate the range of what is possible: ARIMA has been used to project non-communicable-disease-attributable anemia burden using Global Burden of Disease data methodology; the Ghana study projected anemia prevalence among women of reproductive age out to 2030, complete with widening confidence intervals reflecting growing forecast uncertainty further into the future - a feature the client should expect and plan around (a 5-year-out forecast is inherently less certain than a 1-year-out one). A properly time-structured version of our dataset could similarly forecast expected anemia caseload by institution and season, directly informing staffing and supplementation-stock planning.

Why our data cannot currently support this: forecasting is only as trustworthy as the trend and seasonal structure it is built on (9.2, 9.3), since neither can currently be reliably estimated from this dataset, any forecast built on it now would inherit and compound that unreliability, producing a confident-looking but misleading projection. This is a meaningful caution for the client: forecasting from unsuitable data is often worse than not forecasting at all, since a wrong number stated with false precision can drive poor resource decisions more confidently than an honest "we don't know yet."

9.5 ARIMA Modelling

What it is: ARIMA (Autoregressive Integrated Moving Average) combines autoregression (a value predicted from its own recent past values), differencing (removing trend to achieve stationarity), and a moving-average error-smoothing term, conventionally denoted ARIMA(p,d,q); seasonal extensions (SARIMA) add seasonal-lag terms.

Precedent directly relevant to our exact outcome variable: ARIMA has already been used specifically to model and forecast anemia prevalence among women of reproductive age in two separate published studies (Ghana, using ARIMA (1,1,0) fitted to 20 years of consistent annual data with residual diagnostics including the Ljung-Box test for autocorrelation and the Jarque-Bera test for normality; and Iraq, using two decades of consistent national data). Both used long, consistent, single-source time series, the opposite of what our dataset currently offers. The wider

infectious-disease-surveillance literature also documents ARIMA's practical difficulties in real applications (the choices required at each modelling step, and situations where ARIMA modelling fails and alternative approaches such as INAR modelling become necessary), a useful caution against assuming ARIMA is a simple, always-applicable default even where data quality is good. A further caution from recent forecasting literature: ARIMA works best for short-term forecasting when historical trends are stable, linear and non-seasonal, and is less effective when periodic patterns shift suddenly due to interventions or emerging risk factors, relevant here since Sri Lanka's healthcare system has experienced exactly this kind of disruption over the 2014–2023 window our data spans, which would violate ARIMA's stability assumption even if the coverage problem were resolved.

Why ARIMA is not appropriate for our current dataset: ARIMA fundamentally requires a single, regularly spaced, sufficiently long time series. Our data offers neither a consistent institutional source nor regular reporting interval. Attempting to force an ARIMA model onto the current pooled or even single-institution series would produce a model fitted to data-collection artefacts as much as genuine epidemiological signal, precisely the failure mode the literature's own methodological cautions warn against.

9.6 Possible Business Applications (If Suitable Data Were Available)

- **Seasonal staffing and supply planning**: if HB/anemia shows a monsoon-linked seasonal dip (as regional precedent suggests is plausible), the client could pre-position iron supplementation stock and schedule additional screening capacity ahead of the expected seasonal rise in cases.
- **Early warning triggers**: a time series model's residuals (observed vs. expected, given trend and season) could flag an institution or region where anemia prevalence is rising faster than the historical pattern predicts, feeding directly into the Task 10 innovation proposal's early-warning framework concept.
- **Multi-year resource allocation planning**: forecasts with appropriately widening uncertainty bounds (as in the Ghana precedent) would let the client

plan supplementation program budgets 2–3 years ahead with a defensible, quantified range rather than a guess.

- **Evaluating intervention impact**: a sufficiently long, consistent series would let the client formally test whether a future intervention (e.g., a new screening protocol, following Task 6's experimental design discussion) produced a measurable *change* in the underlying trend, not just a before/after snapshot comparison.

9.7 Expected Organizational Benefits

If the data limitations in 9.1 were resolved (i.e., if the client began collecting consistent, continuous, institution-complete monthly data going forward), the organizational benefits of eventually applying time series methods would include: moving from reactive to anticipatory resource planning; a defensible, quantified basis for budgeting supplementation and staffing rather than intuition-based planning; and the ability to detect emerging problems (a sudden rise in a specific institution or age-sex group) against a statistically expected baseline, rather than relying on informal staff observation.

9.8 Conclusion and Recommendation

The single clearest, most actionable recommendation from this task is not a modelling recommendation but a data-collection one: the client should establish consistent, continuous monthly (or at minimum quarterly) FBC reporting across all institutions, with no gaps in institutional participation, as a prerequisite for any future time series work. Only once this is in place would trend analysis, seasonal decomposition, forecasting, and ARIMA modelling become methodologically trustworthy, at which point the published precedents discussed above (particularly the Ghana and Iraq anemia-forecasting studies) provide a clear, directly applicable methodological template to follow. Until then, we recommend the client treat any time-based comparison drawn from the current dataset, including any apparent "trend" a stakeholder might notice in a simple pooled chart, with the same skepticism this task's exploratory analysis demonstrated is warranted.

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